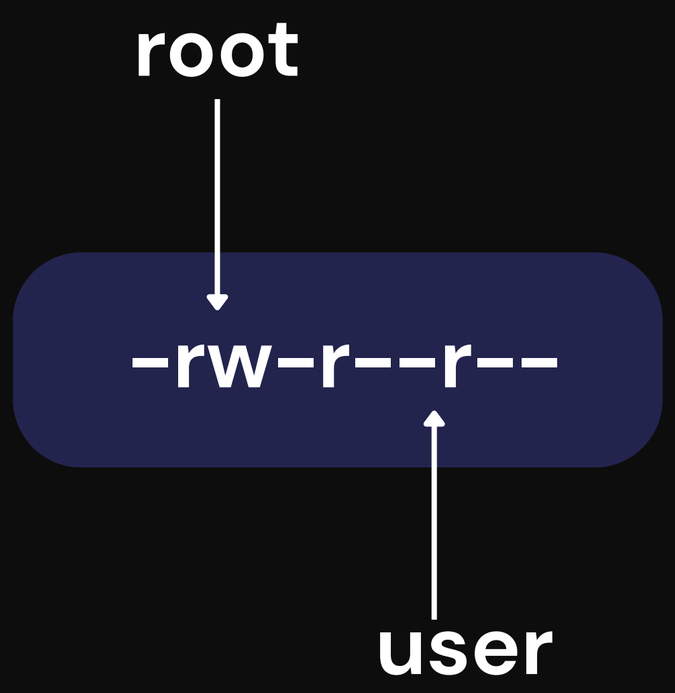
Dirty COW

The Nine-Year Race condition

Saankhya Srikanth

Read only files??

→ Owned by root
→ No write permission



If you're not root, you shouldn't be able to change it

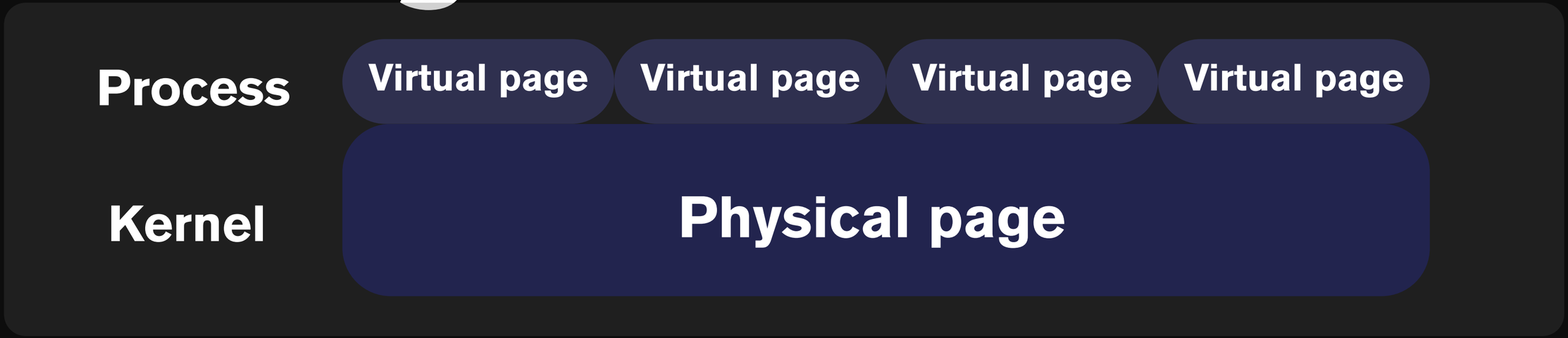
Demo

```
saankhya@fedora:~$ sudo touch file
```

Race condition

Thread 1 gets scheduled first

How does Linux manage its memory?



→ Each process appears to have its own virtual memory
→ Actually optimized internally

Page tables

Virtual Address

↓

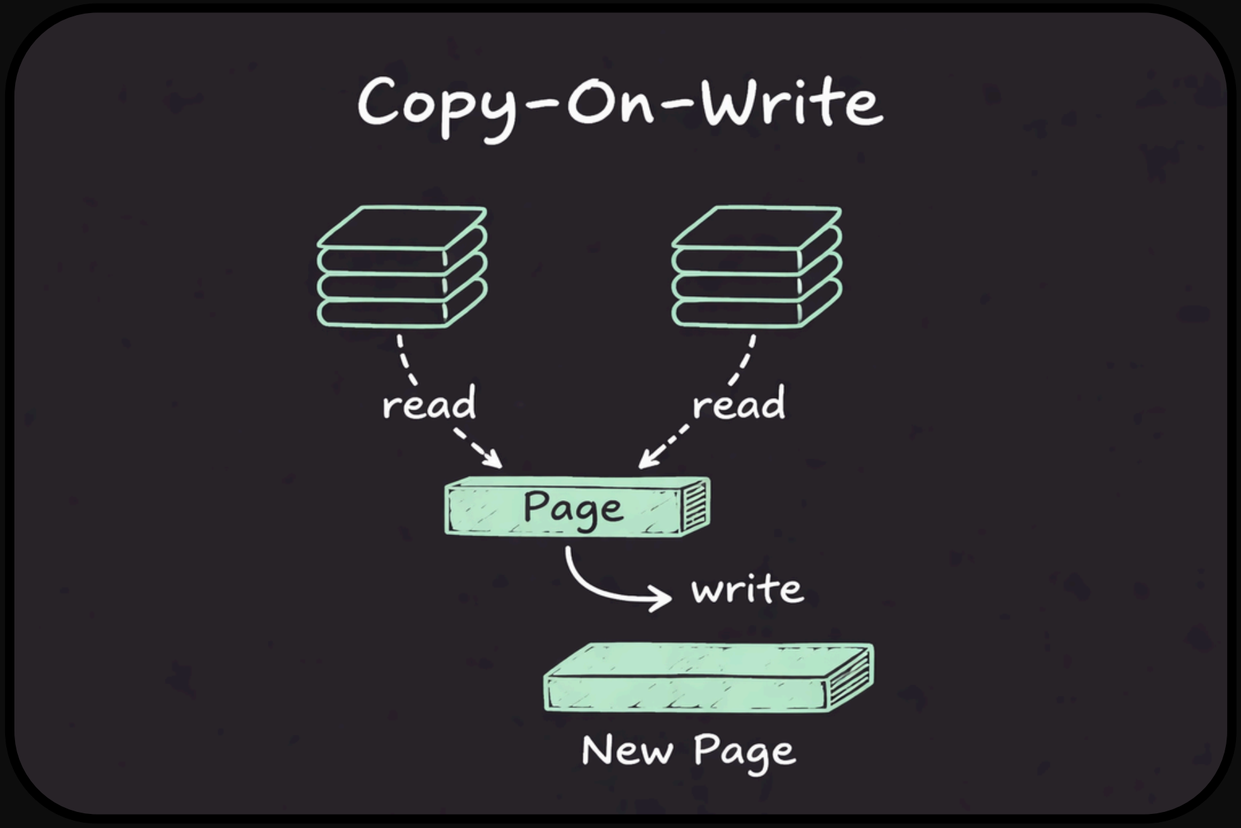
Page Table

↓

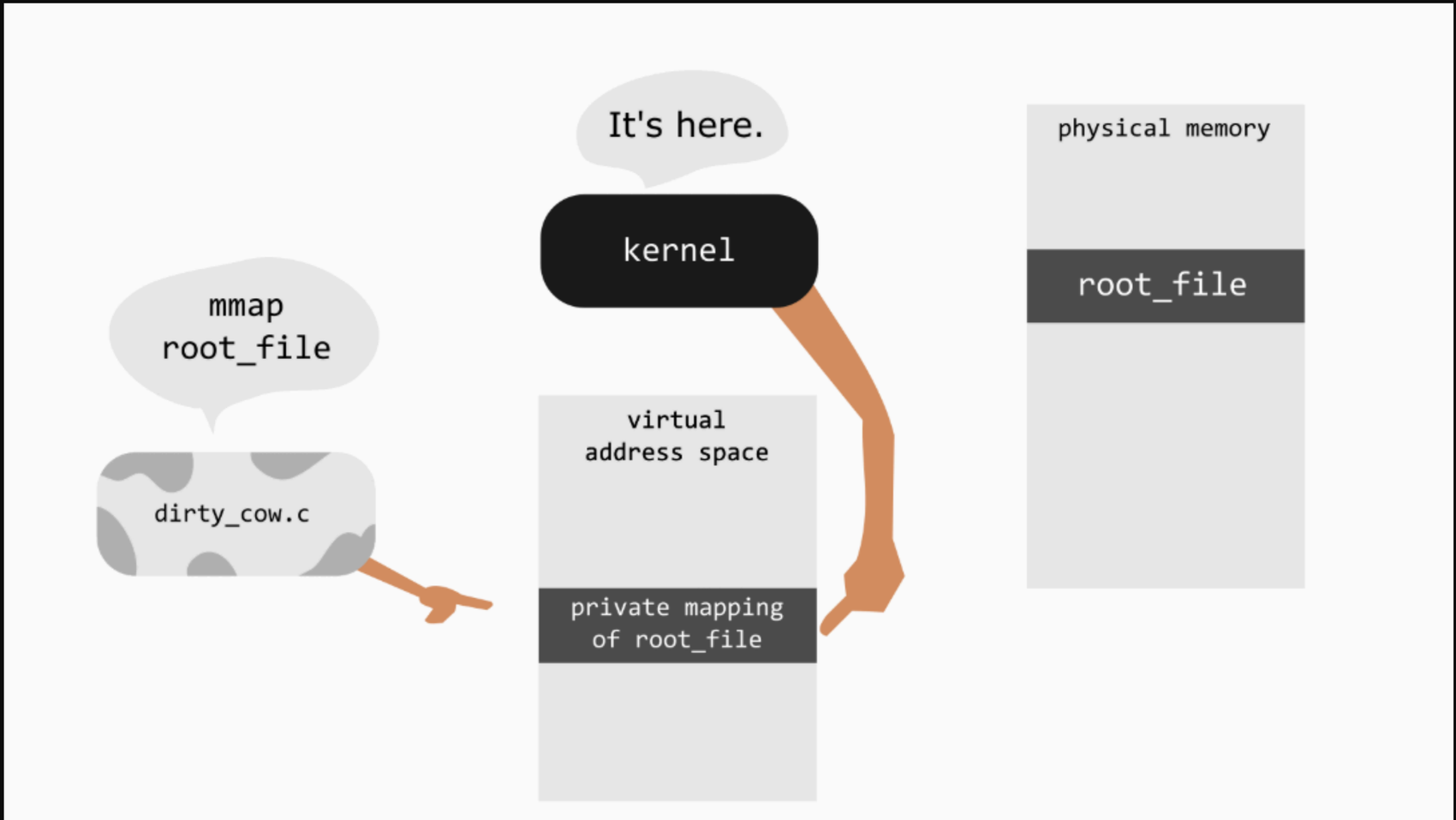
Physical Memory

Copy-On-Write

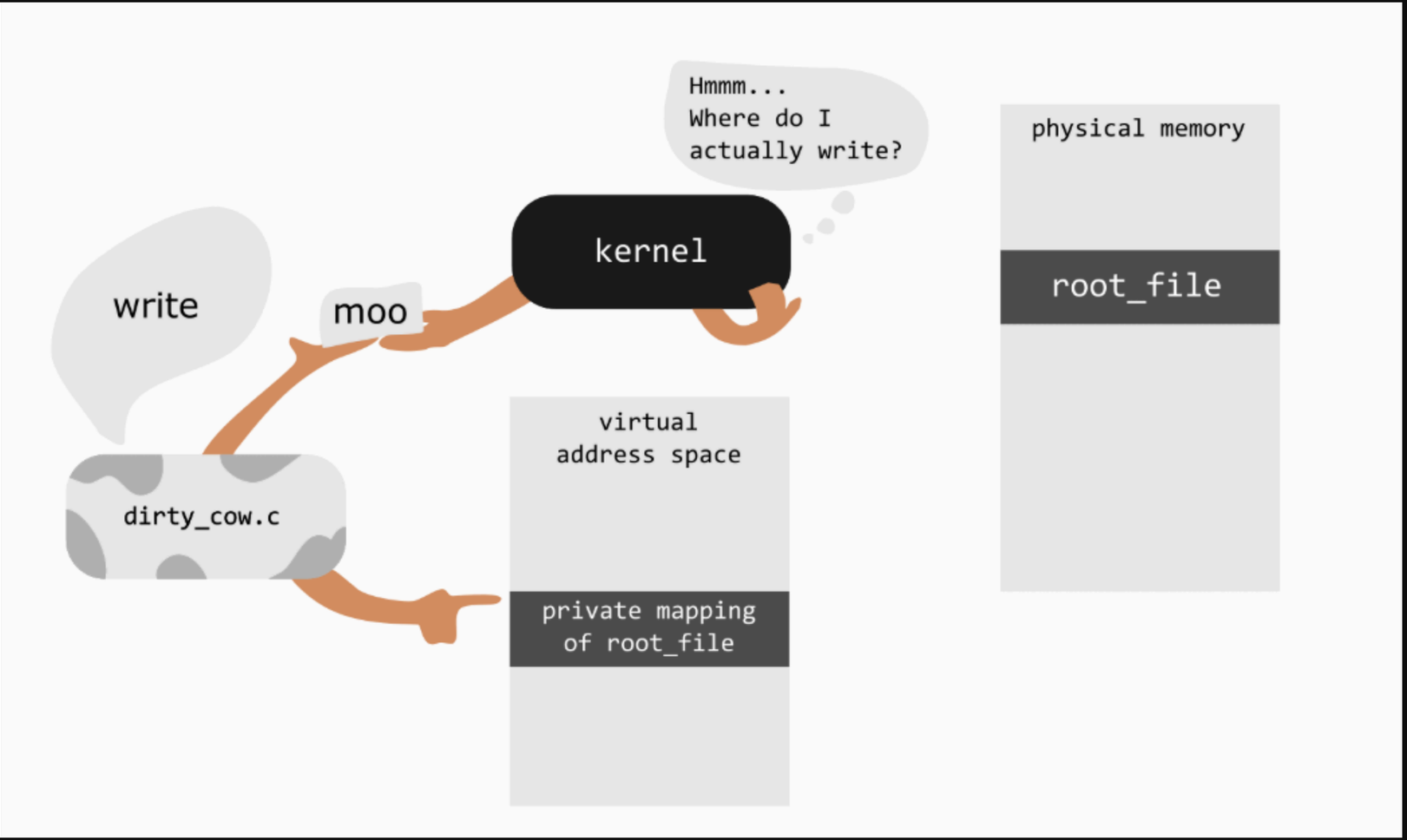
Read? Shared
Write? Makes a copy



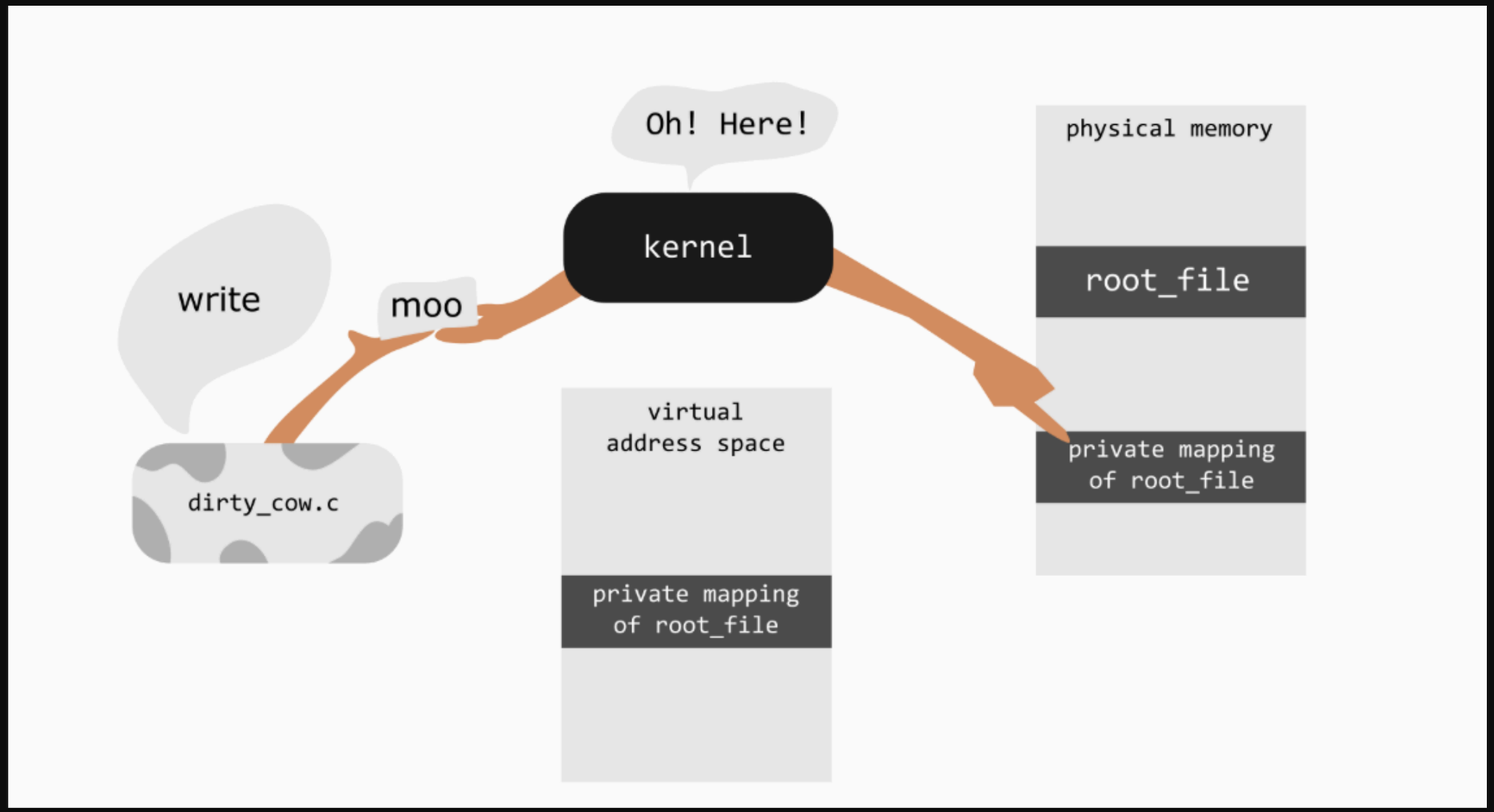
mapping using mmap (no copy yet)
MAP_PRIVATE flag



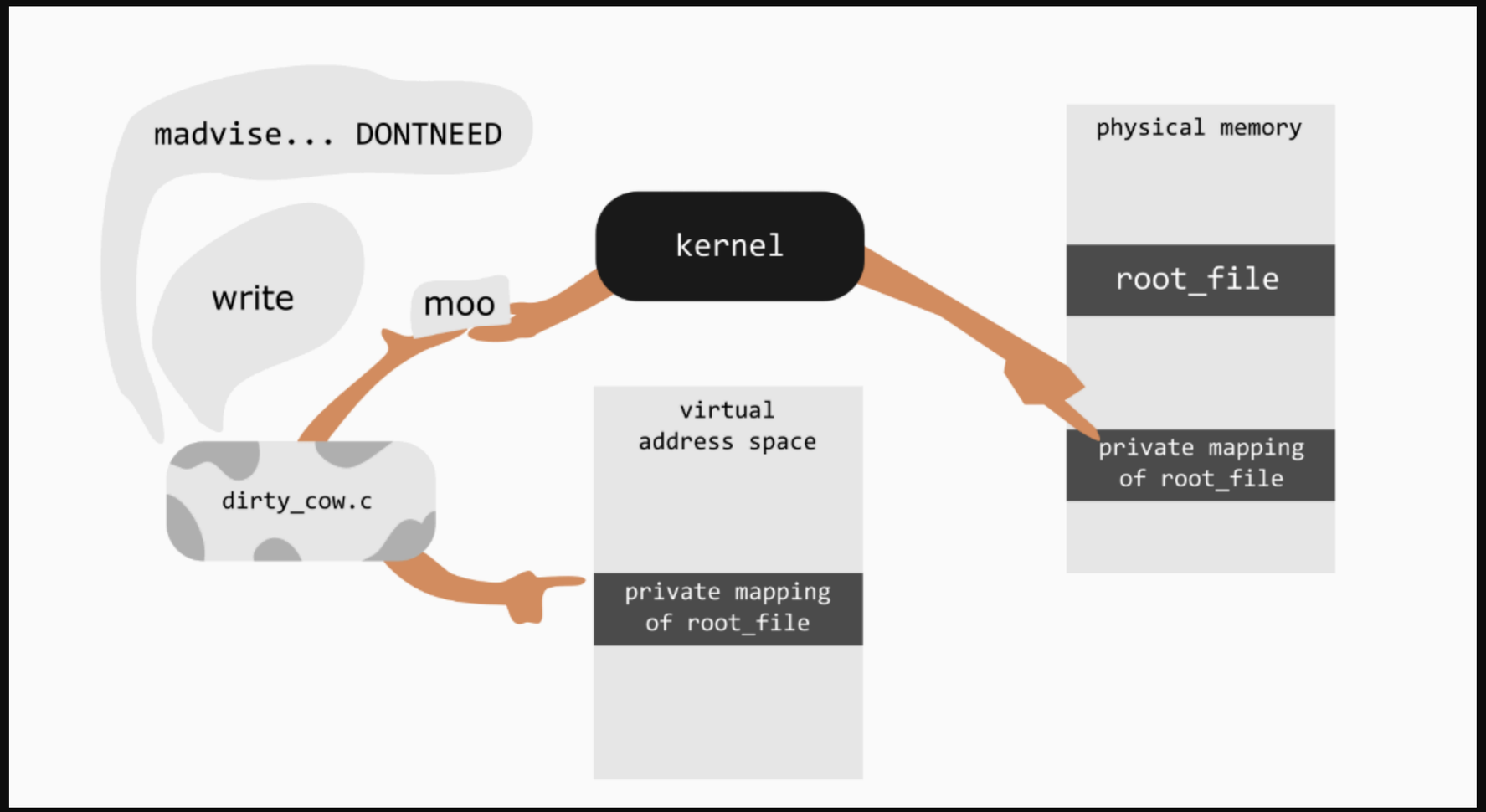
Thread 1 starts write (via /proc/self/mem)



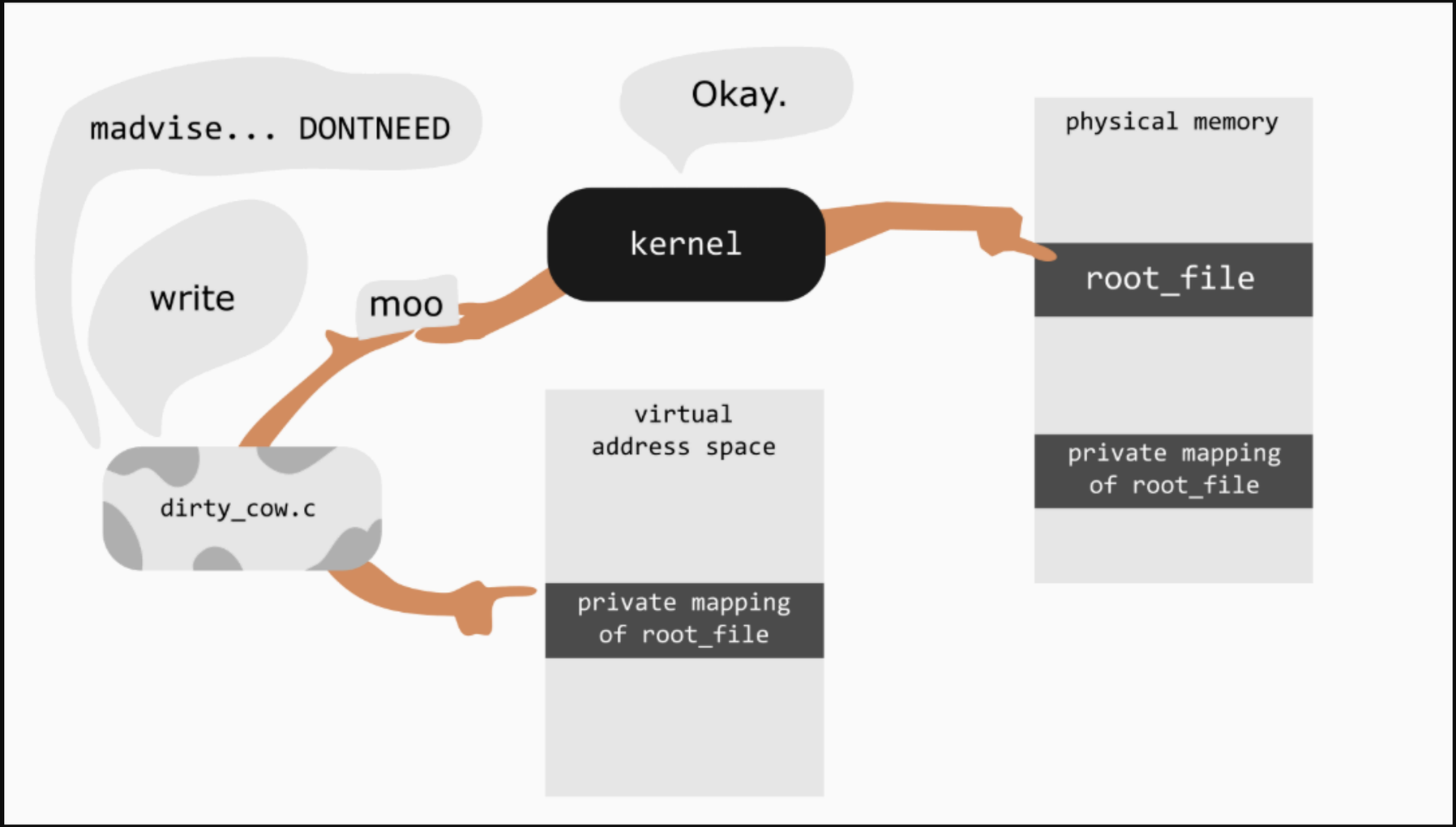
Kernel decides to create private copy FOLL-WRITE



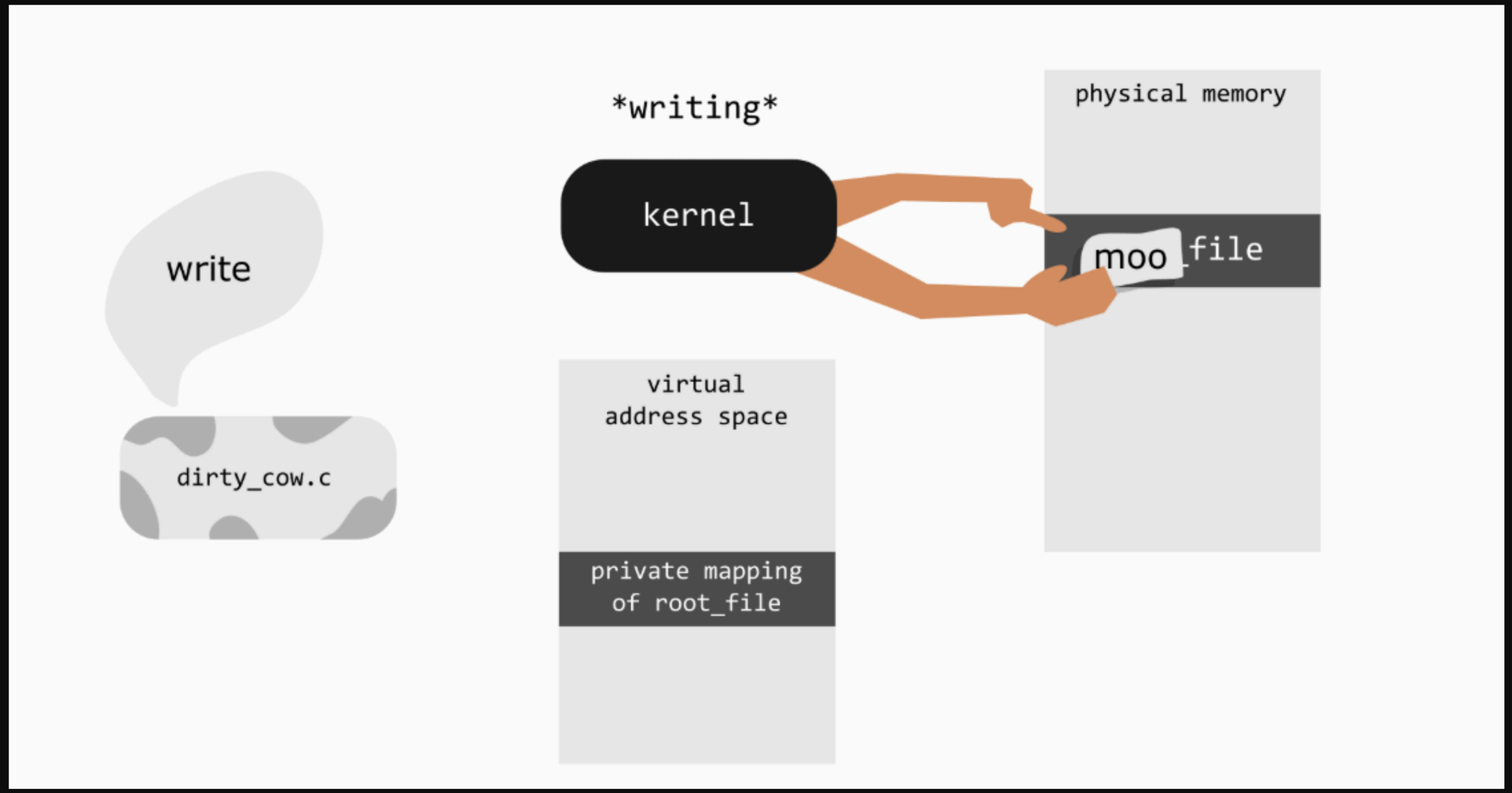
Thread 2 deletes the page (madvise)



Private copy gets invalidated



Writes to the original file



Privilege escalation??

Impact

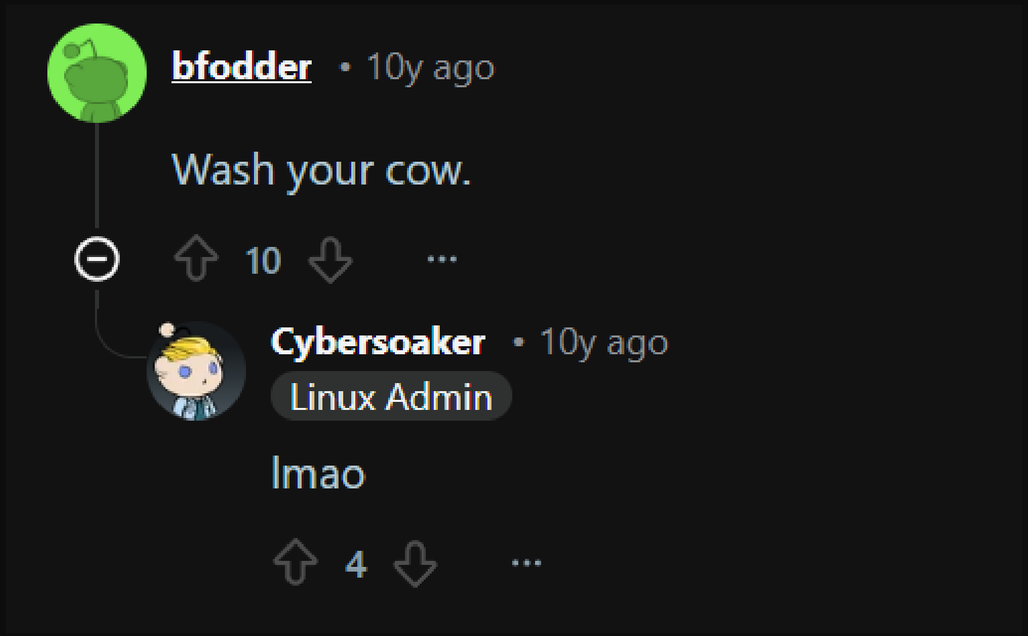
User → Root

Modify protected files

Mitigation

Proper page synchronization

Prevent COW race conditions



iPhone released

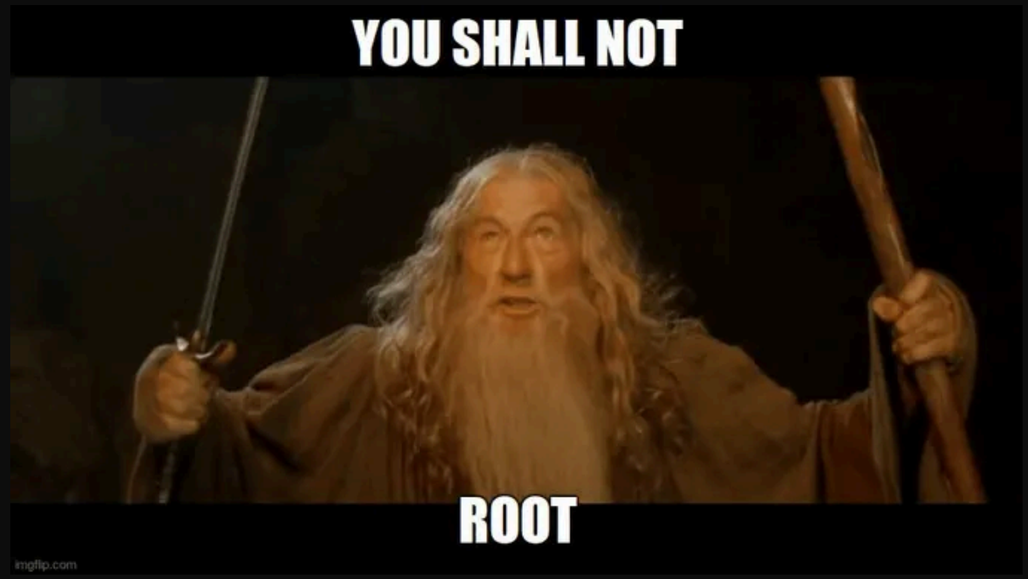
Finally patched

2007 → 2008 → 2016

Android was born

Android exploitation for versions below 7.0

Still relevant ?

Vulnerabilities are patched. Systems stay outdated.

Thank you